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EMBED AFFECT AWARENESS in E-LEARNING: A SYSTEMATIC OUTLINE of the LITERATURE

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Abstract. In recent years, web based education has been perceived as a support tool for instructors as it gives comfort of using it at anytime, any place. Engineering educationalist relies on scientific labeling, like Bloom's classification to make learning goals used to build up lesson's contents. As per Bloom's scientific classification lessons goals can be arranged into three spaces: cognitive, affective and psychomotor area. While evaluating the learners in E-learning, most encouraging courses just put down weight on facts broadcast, to be specific, the cognitive area, and ignore learners learning modalities and feelings due to the distance.

An emotion plays a very noteworthy part in the cognitive procedure of individual, so the change is deficient with no capturing the student's passionate state. With the help of the affective computing technology, it is possible for researchers to understand learner's affect during online learning and then they are able to take corrective action.

This chapter presents an organized review related to affective computing technology. The chapter explores and spots the different modalities, techniques, tools developed and emotions identified by researchers to realize their practicability and development. Finally, few challenges are listed for helping researchers while applying affective computing technology. Total 61 research papers are incorporated in this survey. The result reflects that data is a major concern in this area and very few data sets with some modalities are

only available. Most of the researchers have used facial expression modality for prediction of affective states. Many authors has used generalized term affect while prediction. There is yet a lot of scope in prediction of academic emotions like confusion, engagement, frustration etc. The overall evolution of the research goes through various changes in terms of modalities, technologies, tools and data are targeted.

Keywords. Bloom's scientific classification, E-learning, Affective area, Affective States, Learner's emotions.

1.1 Introduction

Education via the web has opened many doors in the present education scheme. E-learning is fundamentally the framework enabled trade of talent and information by the online instructors to the online students. Web based learning beats the constraint of reality by presenting extravagance of its client to make utilization of it at whenever, anyplace. Similar to usual learning, web based adapting likewise relies upon influential communication of human information, in spite of whether this happens in an eye to eye study hall or over the Internet. An educator depends on scientific classification, like Bloom's scientific classification to make learning goals used to build up teaching contents and exercises. Many times, faculty uses these goals to assess student achievement. It has attractive impact on objectives-based appraisal [6].

Bloom's taxonomy in E-learning helps the instructors to think and analyze their teaching and student's learning. It is used to state clear objectives which can help the teachers to plan lessons accordingly [5]. Bloom's Taxonomy guides the teachers to change the complexity of the questions and helps scholars to acquire greater ranges of hierarchy. Further, it helps to develop critical thinking among teachers. According to Bloom's scientific categorization, lessons goals can be prepared into three spaces: intellectual area, knowledge; Affective area, mentality; and psychomotor area, skills [7]. After the educational strategies are completed, college students must be assessed whether or not they have achieved the presented goals. In the group of students, when the guidance strategy is preparing, the educator always applies several varieties of educating competencies or create inquiries to inspire learners to be thoughtful for accomplishment of affective area instructing goals. The greater part of these aptitudes and approaches are the method of collaboration [3]. E-learning has grown quickly in last few years, and turned into a pattern of learning [4]. For students' assessment, this framework needs to be

planned with the evaluation feature [8]. The widely used evaluation method in distance teaching and learning is on-line examination [8, 9]. Nonetheless, most instructing courses just put down weight on information broadcast, in particular, the knowledge area, and forget about learners learning mentalities and the affective area educating goals. Because of the distance, instructors cannot survey the accomplishment of affective area educating goals. But, if instructors can understand the student's affective condition via the mastering framework, they can defeat the trouble.

As shown in Fig. 1, Emotions have giant influences on learning and has a very noteworthy role in conclusion making and reflecting on the studies. Negative Affect comprises an assortment of bad feelings consisting of outrage, disturb; blame, fear and nervousness can deeply impact training and learning. Bad feelings influence learner's inspiration and learning. It gets hard for student to manage learning exercises and timings. Answer for this is affective computing technology [11]. Consequently, understanding a student's have an effect on all via the learning methodology is critical for getting motivation. In ordinary training research, student motivation can be recognized via questionnaires after occasion. With the development of affective computing, creators can fairly perceive and assess a learner's affective state at the time of learning in a continuous way [14] and thereafter they can appreciate the interrelationship between feeling, inspiration and learning execution.

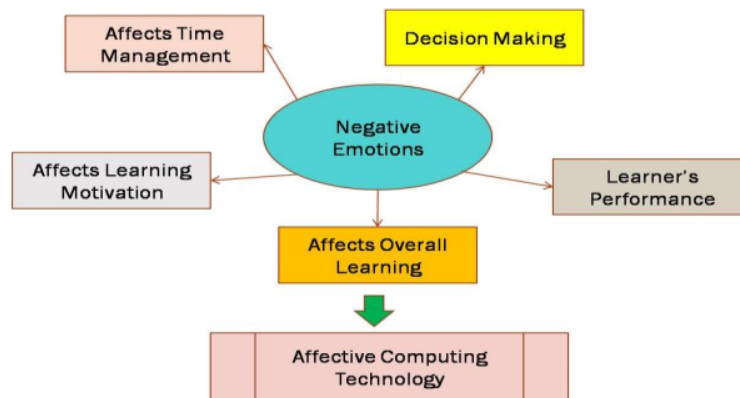


Fig. 1 Role of Emotions in Learning

Therefore, E-courses must place the same importance on the needs of the affective domain along with the cognitive area.

E-learning has been portrayed as less emotion sensitive and greater impersonal due to not acknowledgment of non-verbal communication and gestures of the learner while comparing it to traditional education [9]. On the off chance that internet learning frameworks don't give constant input to the student, at that point student doesn't get the advantage of getting the real comments of mastering performances at some stage in the learning procedure. The student's feelings greatly affect the student's inspiration towards better learning.

With this understanding of importance of affect in E-learning, we are presenting here an organized review of 61 selected papers in this area from 2009 to 2021. These papers are included in the survey after intense scrutiny. Here, we focus on different modalities/instruments, techniques, tools developed and emotions identified by the researchers.

Section 1.2 includes Motivation for this research. Section 1.3 of this chapter represents strategy used in this survey and research questions based on which the survey is designed. Section 1.4 gives the details of related literature included in the survey. Section 1.5 gives survey outcomes and discussion and taken as a whole survey is concluded in Section 1.6.

1.2 Motivation

As shown in Fig. 2, with traditional teaching methods and high understudy/educator proportions, an instructor faces great obstructions in the study hall. Customarily, educators convey the substance and understudies gain knowledge of it. Using this technique, instructors are normally not ready to react to the individual needs of understudies. Besides, because of the huge quantities of understudies in a class, educators cannot concentrate towards every individual understudy. For reviewing they consistently direct a test towards finish of the course. Nonetheless, in spite of the high number of understudies in a study hall, experienced educators typically watch, perceive and tackle the learning way and affective state of the students [31]. The gifted instructors make proper move to emphatically affect knowledge. Be that as it may, the question is the thing that these accomplished instructors "sees" and how they appear at a procedure and whether this movement drives the understudy to a gainful way. Student's learning not only relies on factors like the instructor's teaching style, a

student's capacity and a prerequisite but also depend on the compatibility of emotions.

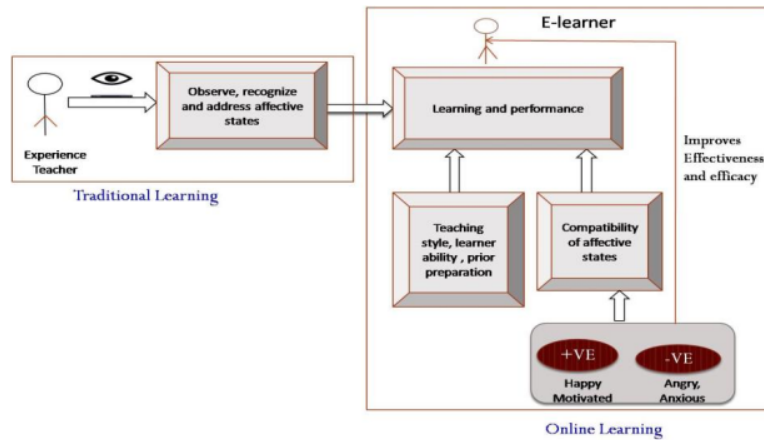


Fig. 2 Research Motivation

The degree to which enthusiastic miracles can meddle with mental existence is no news to educators, understudies who are irate, on edge, or discouraged, experience issues in learning. Individuals trapped in these states don't receive data effectively or manage it very well [1]. Nonetheless, the connection among learning and feelings is a long way from being that straightforward and direct. Good and bad affective states produce distinctive sort of reasoning and this may hold noteworthy inference from learning viewpoint. A reliable theory of discovering that incorporates effectively cognitive and affective factors are robustly required [2].

Gifted educators alter the learning way and their encouraging style as per understudies input signals (which incorporate intellectual, enthusiastic and inspirational perspectives), online learning frameworks mostly don't consider these criticism signals resulting too strict and undermined, as they act in similar way for all understudies. Most of the online learning structures focus on data acquiring or intellectual handling.

While building a framework, affective states are viewed with respect to how the content is organized and introduced. To make

learning effective there is a need to build up a learner's model that coordinates the intellectual cycles and inspirational states [9]. Changing an e-learning framework from non-affect sensitive to a framework that incorporates user's affect requires the representation of a circle called as the affective circle. The affective circle envelops recognition of passionate states, proper activities choice for conclusion making, and the fusion of suitable affect by the framework [8]. Students, who get trapped in bad states like depression don't process and hold information proficiently. Through this, it tends to be construed that a user's affect has a noteworthy role in enhancing the viability of e-learning [10]. With this understanding of affect in E-learning this survey is initiated.

1.3 Research Strategy and Research Questions

The survey related to 'Affective Computing' was carried out in an organized way by following the steps mentioned in Fig. 3. Initially, research questions are designed to provide a suitable path to the review. We take it forward with retrieval of allied research papers and then done scrutiny and selected the most relevant research papers. Finally, the findings of the respective authors are studied and cited along with parameter they have measured in their research.

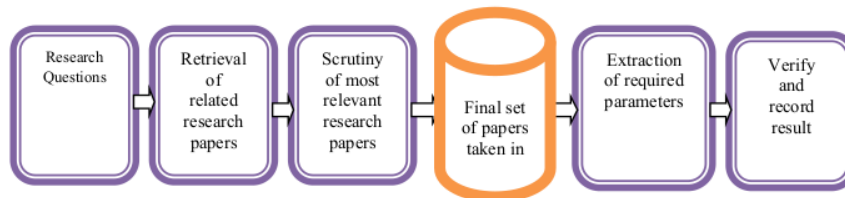


Fig.3 Steps of Strategy used for Literature Survey

To aim the organized review, the following research questions are identified:

- Q1. Which modalities/instruments are used in affective computing study?
- Q2. Which emotions and parameters are measured by the researchers?
- Q3. Which datasets are used by the researchers for prediction of affect?
- Q4. What are challenges facing this research area?

1.4 Literature Review

Emotions and online-learning are connected with one another. The learning execution mirrors, the affective status of online learners in the existing setting of the web based learning in advanced education. There has been a lot of enthusiasm among computer researchers and designers searching for approaches to progress person and workstation interaction by synchronizing emotion and cognition with task imperative [23]. Affective Computing makes the chance of creating computational frameworks for acknowledgment of human affective states and minimize the gap among the profoundly expressive individual and the emotionally challenged computer [23]. Subsequently, Affective Computing research is an interdisciplinary area that plans to investigate human emotional involvement with technology by combining engineering and computer knowledge with areas like brain science, intellectual science, humanism, education and morals [16].

These technologies sense the emotions of an individual by using a mix of various information sources – facial expressions, voice recognition, gestures, sensors [13], eye movement, behavioural log analysis and questionnaires.

Affective computing is a developing field uniting scientists and specialists from different fields, extending from AI, NLP to cognitive and sociologies [20].

The latest couple of years have seen a flood in the utilization of multidisciplinary procedures to consider the difficult role of emotions in a large number of learning settings. Sentiment analyses has picked up fascination in the Computer Based Education research network, maybe due to the interdisciplinary idea of many research teams that include proficiency in brain science computations and NLP.

Emotion information is utilized to improve learning [30]. Here, positive and negative emotions were recognized by utilizing two unique models like behavioural patterns log record and biosensors. Fusion of it is done using Bayesian network [24].

Sandanayake and team have built up a system to know the students online learning performance while estimating the emotional state. Here creators have determined affect from survey and learning performance from standards of conduct in the log record. Few positive and negative emotions were estimated. Critical connection found between the emotions and log record information estimated [17].

Kort has built up an affective model of interaction among emotions and learning.

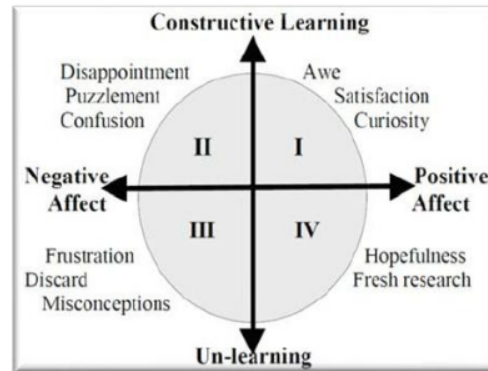


Fig. 4 ¹ Kort's Learning Spiral Model

The flat line is the emotion axis and at right angles is the learning axis. As indicated by Kort in Fig. 4, understudies preferably start with quadrant 1 and may enter into further quadrants while learning where their emotion is bad and have Un-learning state. If understudies get proper support at this point they may move towards hopefulness state [18].

Every student has diverse learning style yet sadly, present E-learning frameworks assess all the students in an equivalent manner. Students learning is rely upon their learning style and affective states. Here authors have recognized students feeling and learning styles (Active or Reflex, Sensitive or Intuitive etc.) from log document. Authors have referred the Fielder Silverman Learning Style Model (FSLSM). As indicated by Fielder Silverman, distinguishing learning styles [11][12][19] are portrayed in Table1.

Table1. Fielder Silverman Learning Styles

Continuum	Preference
Sensing - Intuitive	How you prefer to perceive or take in information
Visual - Verbal	How you prefer information to be presented
Active - Reflective	How you prefer to process information
Sequential - Global	How you prefer to organize and progress toward understanding information

Multimodal combination gives preferable outcomes over the unimodal investigation. There are various sorts of combination strategies utilized for multimodal fusion that are early fusion, late fusion and hybrid fusion. Finding summed up model and utilization of deep learning for multimodal fusion is significant future work in this area [20].

Affective PC not only gives improved execution in helping individuals, but also improves PC's capacities to decide. In view of this proof, to construct PCs that make intelligent decisions may require construction of computers that have emotions [16].

In online learning environment, feelings majorly affect basic psychological cycles; neurological evidence shows they are not a luxury. Here authors have featured a few outcomes from the neurological literature which demonstrate that emotions play a fundamental job in human imagination and knowledge, however furthermore in critical human thinking and conclusion-making. PCs that will collaborate as expected and intelligently with people require recognizing and expressing affect [15].

Some experiment took data from biosensor wearable gadgets like smart watches and fit band. Authors have utilized here emotionWear Framework. In emotionWear framework a user watches the contents in the Smartphone which incorporates VR headset and it is associated with the cloud storage. The user wears remote detecting gloves which produce various signs which are utilized to recognize emotions of the user. The emotions detected were exact yet there was delay in investigation [27].

Most of the work aims to discover affect using facial expressions and sensors. The overall purpose is to enhance accuracy and take some corrective action to enhance learning. Researchers have performed experiment on data collected at various universities and labs. Different tools were developed by the researchers for data collection and analysis. Table 2 summarizes the parameters measured, modalities used, methodology and final outcome of the papers included in the survey.

Table 2: summary of the parameters measured, modalities used, methodology and final outcome of the papers

Sr. No.	References	Parameters Measured	Modality/ Instrument	Methodology	Benefit/Outcome
1	Liping Shen [32] 2009	Interest, Boredom, Confusion, Satisfaction Frustration, Engagement Hopefulness, and Disappointment	User Profiles and Learning Events and Biosensors are used.	Study was guided by Russel's Circumplex Model and Korth model. Authors explored evolution of emotions at the time of learning and observed use of emotion feedback to improve learning.	Experiment confirms performance of emotion aware module over non-emotion-aware is more.
2	Farman Ali Khan [33] 2009	i. Emotions: Confidence, Confusion, Independence and Effort ii. Learning Styles	Behavioural Log File: Consist of 16 factors	Both Emotion and Learning Styles are identified from Log file.	Tool is presented for detection and calculation of various styles of learning and affect.
3	Farman Ali Khan [34] 2010	i. Emotions: Confidence, Confusion, Independence and Effort ii. Learning Styles	Behavioural Log File: Consist of 16	i. Both Emotion and Learning Styles are identified from Log file. ii. Adaptive course generator is used. iii. Adaptive Affective tactic generator module is developed which will do tagging and provide help according to LS and AS.	The investigation of learner's styles of learning and affect is done for providing them personalized support.

4	T.C.Sandanayake [17] 2013	Few ¹ positive emotions: Pride, Hope, Enjoyment & Relief and some negative emotions: Anger, Boredom, Anxiety, Hopelessness and Shame	i. Behavioural Log File: Evaluating Learners Performance ii. Questionnaires: Measured Learners Emotions.	Tool is developed to recognize E-learner's emotions. Multiple regression is used.	i. Identified learner's learning level and the emotional state. ii. Correlation between learners observed behaviour and emotions is analysed.
5	NELL BUISSINK-SMITH [3] 2014	Affective attributes	Review of primary and secondary source data collection methods are given.	Overview of affective learning in education is given	Framework for measurement of affective attributes is proposed.
6	M. Feidakis[35] 2014	Anger, boredom, confusion, fatigue, relief/relaxation, curiosity, despair, excitement, disinterest, embarrassment, inspiration, interest, stress and nothing	Tool Emotcontrol is developed. Self reporting is used.	i. Authors has developed conceptual and computational model. ii. Implemented model employs self-reporting of affect, feedback and efficient emotion visualisations.	Users found Emotcontrol tool useful, expressive, usable and effective.
7	Chih-Hung Wu [14] 2015	Emotions	Details of physiological signals, multimodal based approach and other modalities are given	For realization of latest trends, here authors have summarized ² measurements of affective computing in learning field.	i. The trends of Affective Computing in learning are discussed. ii. Multimodal framework for affect recognition is proposed. iii. Challenges and problems in this field are proposed.

8	Rachel Carlos [36] 2015	Affective states in collaborative environment	Review Paper	Literature survey is done to find i. Frequently used emotional states in collaborative surroundings ii. Were affective states empirically evaluated? iii Types of surroundings which makes use of affective states to support group learning	Authors verified that: 54,84% studies deal with emotions' 51,61% studies were empirically evaluated and LMS and E-Learning systems are the surroundings which makes use of affective states to support group learning.
9	Manuel Rodrigues [37] 2015	Affective states and Learning styles	Only framework is given where authors may use Explicit assumption: Self reporting and Implicit assumption: Facial Agent, Keyboard agent, Mouse agent, Log agent.	Authors proposed framework in which after linking module with Moodle platform, student's affective states and learning styles get detected and contents were presented accordingly.	i. In this paper, some problems of Moodle are identified. ii. The framework is proposed here which includes an affective module in Moodle.
10	Jason Matthew Harley [38] 2015	Review of emotions in computer based learning	Pro's and Con's of i. self reporting ii. Log files iii. Facial expressions iv. Language and Vocalisation v. Psychological signals are given.	Survey of all methods and modalities is given. The effectiveness of multimodal approaches is discussed.	Multiple methods used to predict emotions with their pros and cons are given.

11	Arvid Frydenlund [39] 2015	Affect, dominance, favourability, valence, familiarity.	Multimodal affect detection is done. Sensors and videos are used.	Here, authors have presented a multimodal system for affect prediction. They have used DEAP data set to train model. They also have introduced a way for extracting video features and creating more training examples from data.	Proposed model performs better for affect, dominance, favourability and valence than baseline methods.
12	Jyotish Kumar [40] 2015	Disgust, Anger, Love, Courage	Galvanic skin response (GSR) physiological signal is used to measure electro- dermal responses.	Here, GSR of participants was recorded by placing skin electrodes at the palmer surface while watching videos. By analyzing this data, emotions are detected.	Authors have confirmed the performance by comparing it with verbal reported feedback.
13	Ashwin T S [41] 2015	Basic emotions	Facial expressions detecting multiuser faces in E-learning	Here authors have implemented multiuser face detection based e- Learning system using SVM. The method gives 89% to 90% accuracy using different datasets LFW, FDDB, and YFD.	The method proposed by authors recognizes emotions for several faces in a solo frame.
14	Nigel Bosch[42] 2015	Boredom, confusion, delight, engagement and frustration	Facial expressions	Here, authors proposed a method to detect students' affect. Data were collected in the computer lab. Affect is detected using facial expressions and body movements.	Here, authors has validated face- based detector for learning- centred affect. They have proved, it was possible to detect affect in the wild even though challenges exist.

15	Ishan Behoora [43] 2015	Engagement/ Interest, Delight, Frustration, Boredom	Non wearable sensors like the Microsoft Kinect which captures physical image and motion	Authors have developed a system. Steps of it are as follows. i. Data acquisition and feature generation ii. Detecting body language from these features and mapping emotional state of design team members. iii. Machine learning modal is trained.	Emotional states are detected with accuracies above 98%.
16	Abhay Gupta [21] 2016	Engagement, frustration, confusion and boredom	Video analysis	Authors present DAiSEE dataset which includes video's of academic emotions.	DAiSEE free to use, crowd sourced dataset for modelling four affective states engagement, frustration, confusion and boredom in e-learning environments.
17	Ramanathan Subramanian[22], 2016	Affect and Personality traits	Physiological signals: GSR, Frontal EEG, ECG and facial features	Authors present dataset which includes 58 participants and 36 Videos. Data is collected using commercial wearable sensors and a webcam.	Authors present novel multimodal affective database ASCERTAIN.

18	Qianyu Hu [44] 2016	Positive, Negative, Joy, Anger, Pleasure, Irritation, Focused, Shame, Interested, Guilt, Calm, Anxiety, Peaceful, Stress, Comfortable, Frustrated, Attentive, Distracted and Bored	Questionnaires	i. Vary the difficulty of the task. ii. Perceived affective states and performance time. iii. Correlation test and performance analysis is done.	i. No strong correlation between performance time and perceived affective states showed in a simpler task as opposed to some correlations in a more complex task. ii. Relationship is provided by authors to improve the laboratory performance and safety.
19	Nik Thompson [45] 2016	Enjoyment, activation and valence	i. Questionnaire involved during measurement of perceived learning and for enjoyment. ii. Sensors are applied to understand activation and valence.	Here, authors have developed Affective Tutoring System and compared it with a non affective version. Affective version show measurable improvements in perceived learning.	Affective tutoring system is more effective than non affective tutoring system.
20	Bahreini, Kiavash [46] 2016	Neutral, fear, sad, happy, disgust, surprise, anger	Vocal and facial expression recognition	Real time vocal and facial expressions are captured through videos and analysis is done. Feedback is provided.	Here, developed software allows unobtrusive observation of learners' behaviours and converts it into emotional states.

21	Ray, Arindam [47] 2016	Happiness, Sadness, Surprise, Anger, Disgust, Fear,	Sensors and facial expression	Here, authors proposed Affective Computing Module in E-Learning system. They explore evolution of emotions. Based on the detected emotion, the system delivers lesson using Fuzzy Logic & Artificial Neural Network.	Achieve more accuracy at decision level fusion of facial expression and sensor data.
22	Akputu K. Oryina [48] 2016	Learning emotions Hopefulness, engagement, Happiness, Frustration, Surprise, Boredom, and Confusion.	Facial Expressions	Here, author's present new architecture Facial Emotion Recognition Technology (FERT) having three modules, face detection, feature extraction and emotion recognition.	This paper presents a new architecture FERT. Author has verified Multiple Kernel Learning (MKL) framework which outperforms traditional classifiers by conducting experiments on contextual emotion datasets.
23	K A Laksitowening [49] 2017	Identified: Learning type from 3 factors Learning Styles, Motivation and Knowledge Ability	Behavioural Log File	Framework of triple factor approach is provided and illustration of dynamic personalization is given.	Here authors have used dynamic personalization in which contents are given to learner based on result of triple factor approach.
24	Christian E. Lopez [50] 2017	Performance of learner	Non-wearable infrared sensors like camera are used.	Here SVM is used to classify the one's performance based on facial key point data which is captured using a non-wearable sensor. Cross validation approach is used to evaluate the accuracy of model.	Authors have designed system for providing feedback to individual based on task and characteristics of user.

25	Shelena Soosay [51] 2017	Psychomotor skills and affective states.	Scores given by evaluators	Here, authors have prepared an assessment rubric for evaluating different projects based on the psychomotor and affective learning domains aligned with Bloom's Taxonomy.	Rubrics designed here is helpful for the students while developing a software project.
26	Tara J. Brigham [13] 2017	Affective states: Emotions	Sensors were used	Affective computing technology collects data using different sensors and analyzes it to get emotional state. Then system give response based on detected state. This technology is very helpful into future.	While making use of affective computing technology, ethics, privacy and cost-effectiveness are major concerns.
27	Soujanya Poria a [20] 2018	Affect	Multimodal Affect: audio, visual and text information	Here, authors had given systematic review of multimodal affect frameworks and different fusion techniques.	This survey has confirmed that multimodal classifiers outperforms than unimodal. Text modality is more important and deep learning approaches are used most widely for extracting features.
28	Jan K. Argasi'nski [52] 2018	Creation of Serious Affective Game	Design Patterns	Authors proposed a novel framework for the designing and assessing affective games based on design patterns.	Authors have proposed a novel approach for designing affective serious games.

29	Chih-Hung Wu [53] 2018	Affective states: angry, fear, happy neutral, contempt, disgust, sad, and surprise.	Multimodal: facial expressions, heart rate monitoring, blood oxygen level, skin conductance response, and electroencephalogram signals.	Here, authors have developed the data integration system for integrating data of multiple modalities. Real time facial expressions are classified into six basic emotions.	Authors have developed integration mechanism for avoiding sampling problem. It average out sensor's data and rearrange it into same time. System is useful for collecting and integrating multimodal data in affective computing field.
30	Rwitaajt Majumdar [54] 2018	Engagement	Log file in TPS architecture	Tool is developed to collect data using observational protocol and to analyse results. Analysis and tracking of collected data with respect to time and state of change is done.	Tool is helpful to Instructor and researchers.
31	Francesca D'Erricoa [55] 2018	11 cognitive emotions like interest, attention, concentration, surprise, disappoint, frustration etc.	Facial expression	Here authors has collected data in the form of video's of 10 psychology students during online learning task and using software detected the emotions.	Authors tested the reliability of software developed for classification of cognitive emotions based on facial expressions.

32	Siddharth [56] 2019	emotion/ valence/ arousal/ liking	Sensors and Video data	Here authors have applied novel deep-learning based methods on publicly available datasets. Authors have done experiment on each separate dataset as well as by doing fusion of features. They proved that their algorithms outperform for classification on DEAP and MAHNOB-HCI datasets.	Authors proposed here a novel technique for identifying brain sections corresponding to various affective states.
33	Yinghui Zhou [57] 2020	Emotional state and Learning state	2 Expressions, speech, eye movement, physiological signals, text, questionnaires and multi-modal.	Here, authors proposed a framework 2 for online learning supported by Artificial Intelligence, Big Data and Brain Computer Interface. Personalized education was also provided by the system.	The framework proposed helped to obtain more real data. It makes efficient analysis and recognises emotions more accurately.
34	Troussas C [58] 2020	Affective states	Sentiment analysis	Here author 6's focuses on the concepts of affective computing adapted to social networking-based learning and learners' affective states.	6 authors concluded sentiment analysis refers to the use 10 expert methods to systematically identify, extract, quantify and study affective states and subjective information.

35	Silvia Ceccacci [59] 2021	Emotion	Facial coding techniques	Here, authors have developed a system which takes as input the video capture by the webcam of the device used to attend the course. i. It Perform continuous students' authentication based on face recognition ii. Monitor the students' level of attention through head orientation tracking and gaze detection analysis. iii. Estimate student's emotion during the course attendance.	Authors describe the overall system design and report the results of a preliminary survey, which involved a total of 14 subjects. Intension of authors is to check user acceptance to continue using such a system
36	Resham Arya [60] 2021	Affect	Sensors, Facial Expressions	A survey of domains contributing to affective computing is given in detail.	Here, Authors has explained different domains in Affective computing, presented few existing affective databases, applications are discussed and some challenges along with suggested solutions were given for future work
37	Saurabh Kumar [61] 2021	Affect	Text and image	Here, authors presents affective computing model using both text and image. Experiment was conducted on standard datasets using deep learning models.	The result confirms that the proposed method outperforms the earlier methods.

From the literature survey, it is discovered that various modalities like facial expression recognition, questionnaires, interaction log behaviour, eye movement, gestures, voice recognition and sensors (Galvanic Skin Response (GSR), Electrocardiogram (ECG) and Electroencephalography (EEG) etc.) are used by researchers to recognize the emotions.

From the literature survey done it was revealed that, different researchers have developed the tool, emotions were predicted and the criticism is given to the learner. Due to criticism the improvements are observed in the learner's performance.

1.5 Survey outcome and Discussion

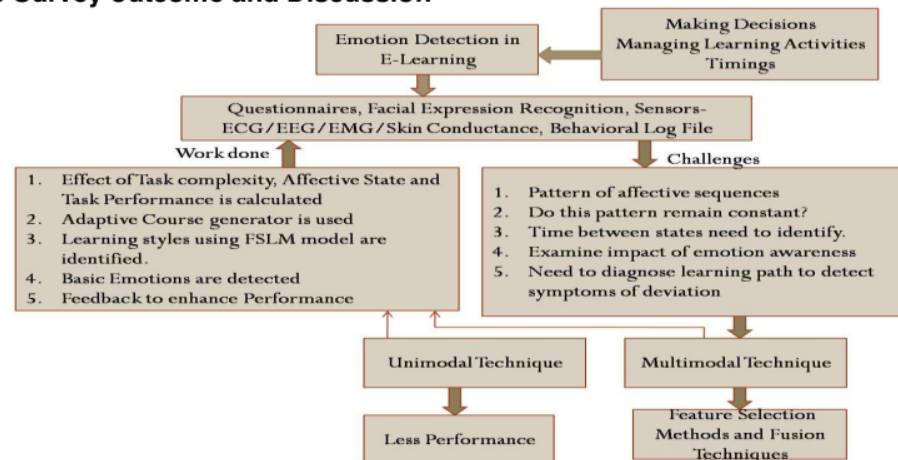


Fig. 5 Literature Analysis

As shown in Fig. 5, emotions have an enormous role in E-learning. Work previously done in the field of affective computing by various authors is as follows.

- Effect of Task complexity on affective state and task execution is identified by giving the mechanical job of cutting piece in rectangular and spherical structure to students.
- According to the learning styles and emotions of learner, course contents will be given to them to upgrade their performance.
- Learning styles of learners were recognized.

- Fundamental emotions were recognized by individual researchers.
- By analyzing the performance based on historical data, feedback is given to learners. Improvement in performance is observed using this method.

From the literature survey it is discovered that various modalities like facial expression recognition, questionnaires, interaction behaviour log, eye tracking, gestures/body motions, language and vocalization, text and sensors (Galvanic Skin Response (GSR), Electrocardiogram (ECG) and Electroencephalography (EEG) etc.) are used by researchers to recognize the emotions.

There are two different techniques unimodal and multimodal technique used by researchers to detect emotions in E-learning. There are different feature selection methods and fusion techniques used in multimodal technique. Different fusion techniques are **early fusion** (fusion at feature level), **late fusion** (fusion at output level) and **hybrid fusion** (mixture of early and late fusion). Performance of multimodal techniques is better than unimodal technique.

From this survey, it has been observed that most of the researchers have used facial expressions to recognize the emotion which is followed by sensors modality as shown in Fig. 6.

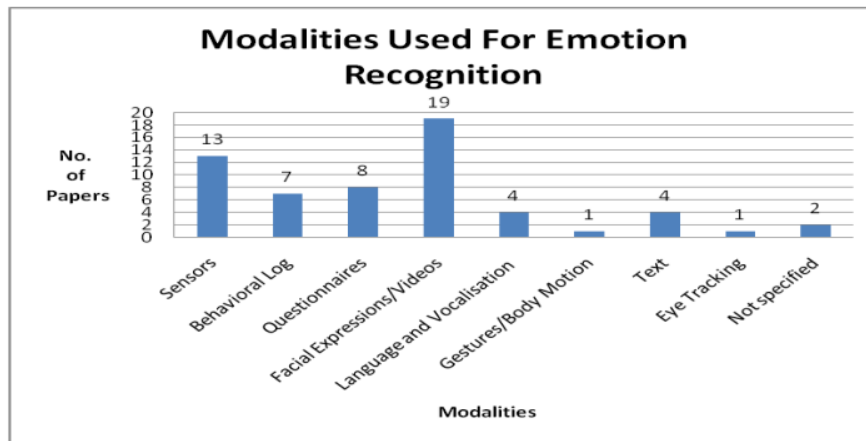


Fig. 6 Modalities Used for Emotion Recognition

From the literature survey done it was revealed that, researchers have predicted different emotions, developed the tools and the criticism is given to the learner. Due to criticism the improvements are observed in the learner's performance. Most of the researchers have worked on academic emotions like Interest, engagement, confusion, frustration, boredom and hopefulness. Affect is generalized term used by the researchers, found in most of the papers.

Fig. 7 shows instruments/modalities used and parameters identified by researchers in the field of affective computing. After identifying parameters researchers has taken various actions like use of adaptive course generator, dynamic personalization, giving feedback based on state.

Instrument used	Parameter		Action
	Enjoyment	Attentive	
Questionnaire	Pride	Fatigue	Learning Companion
Log File	Hope	Irritation	
Sensors	Relief	Guilt	
Facial Expression	Anger	Stress	Adaptive course generator is used
Videos	Anxiety	Distracted	
User Profiles	Hopelessness	Activation	
Focus group interviews, observations	Shame	Valence	Co-relation between affective state & personality trait
Microsoft connect camera depth sensor & micro phone capture, physical image & motion	Boredom	Excitement	
	Interest	Inspiration	
	Engagement	Disgust	Dynamic Personalisation
Facial agent, keyboard agent, mouse & log agent	Confusion	Fear	
	Frustration	Neutral	
Language & vocalisation, audio + text	Satisfaction	Sad	Personalised feedback based on task & unique characteristics of individual
	Disappointment	Surprise	
	Confidence	Familiarity	
Multimodel	Independence	Emotion	Learning Dashboard
Body part gestures	Effort	Value, Attitude, Behavior	
Social Interaction analysis (tweets, comments etc.)	Focused	Motivation	
	Calm	Commitment	Feedback based on state
	Peaceful	Self esteem	
Learning analytics	Comfortable	Curiosity	
API is used			

Fig.7 Instruments Used, Parameters Measured and Action taken by
Researchers in Affective Computing Area

There are various challenges need to address in this field.

1. Instead of only finding affective state, there is a need to identify “What are the patterns of affective sequences of E-learner during E-learning?”
2. “Do these patterns remain constant throughout the course?”
3. Time span between two different emotions need to identify.
4. Need to examine result of emotion awareness.
5. Need to diagnose learning path to detect symptoms of deviation.
6. Data for few modalities is a major concern.

As given in Table 3, there are few affective datasets which are available for affect recognition. But for certain modalities like interaction log, data is a major concern.

Table 3 Affective Databases

Sr. No.	Dataset	Modalities	Affect	Purpose
1	DEAP	Physiological	Yes	Implicit affective tagging from physiological signals
2	ASCERTAIN	Face, physiological	Yes	Affect and personality recognition
3	AMIGOS	Audio, visual, physiological, depth	Yes	Personality, mood and affect recognition
4	DAiSEE	Video	Yes	Data set for affective states in E-Environments
5	DECAF	Face, physiological	Yes	Affect recognition
6	MANHOB-HCI	Face, audio, physiological, eye gaze	Yes	Emotion recognition

1.6 Conclusion

At present, emotion recognition can be carried out by using various modalities. Among them, facial expression and sensors are more widely used for emotion recognition, but emotion recognition based on gestures/body motions and eye tracking are less applied. This study has looked into the latest trends in Affective Computing in teaching and learning. In this chapter authors have discussed about

what is the work already done in this area and what challenges need to be addressed. Specifically, we featured major studies in affective state recognition, which we believe vital segments of any affect detector system. Our study has affirmed other researcher's discoveries that multimodal classifiers can broadly get utilized and beats unimodal classifiers. As acknowledged in this review, a portion of the key exceptional difficulties in this energizing field include: 1) Data collection 2) Finding patterns of affective sequences. 3) Do this pattern remain constant throughout learning? 4) Does the learning style influence the exhibition of student? and so on. These difficulties propose we are still a long way from creating a constant affective state detector which can successfully and emotionally speak with people and feel our emotions. Data for certain modalities is a major concern in the field of affective computing.

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